

UWB Is A SUPERIOR WIRELESS TECHNOLOGY In Terms Of Bringing Spatial Context For WIRELESS DEVICES

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Although the ultra-wideband (UWB) technology has existed for decades, its evolution took centre stage recently with the growth of the Internet of Things (IoT). It is considered to be one of the youngest cousins of the wireless tech family. Sanjay Gupta, vice president and India country manager, NXP Semiconductors, sheds light on the technology's potential in an interaction with Ayushee Sharma

Q. How is UWB different from other wireless technologies?

A. Ultra-wideband (UWB) is a short-range wireless communication protocol that can provide precise, secure, and real-time localisation capabilities unrivalled by other wireless technologies such as Wi-Fi, Bluetooth, NFC, and GPS. One can transmit large amounts of digital data over a wide spectrum of frequency bands for comparatively shorter distances. Data pieces can be carried over a distance of up to 71 metres at a very low power of not more than 0.5 milliwatts. But these have the capacity to conquer obstacles in their way that appear to reflect the signals.

UWB can also be leveraged for enhanced security access with localisation. It transmits pulse at specific time intervals with which systems can further use the transmitted information to determine the time-of-flight and directional information. As a result of transmission in specific time intervals, it prevents the interception and rebroadcasting of the information in what is referred to as man-in-the-middle attacks where further encoding the information makes the solution seemingly impregnable to attacks.

The competitive edge over its predecessors is that UWB is a superior wireless technology in terms of bringing spatial context for wireless devices. Unlike GPS, which is known for its outdoor localisation capabilities, UWB further extends its services to both indoor and outdoor environment accurately; whether it is finding out where we parked our car, allowing users to unlock doors (with the key fob), a device to device communication (locating a particular person), pinpointing other UWB-enabled devices, or providing secure hands-free experience to users.

Q. What are the applications of UWB?

A. UWB is believed to be one of the most secure wireless means to connect different home electronic systems and other respective IoT-enabled smart devices such as televisions, audio speakers, camcorders, secure building access (doors and garages), lights, etc.

Today, UWB has transformed from just a data transmission technology to secure ranging technology, having multiple use cases for numerous applications. Some use cases of UWB are—secure wireless payments (considered securer than NFC), smart retail and asset tracking (through smart UWB tags), wearable health sensors along with tracking and monitoring of objects as well as people. Last year, NXP collaborated with BMW and Continental to enable a global standard for hands-free smart access with precise, secure, and real-time localisation capabilities based on UWB chip, which can transform smartphones into digital and secure car keys.

Q. What would UWB integration mean for smartphones?

A. The technology is sufficiently accurate to enable completely hands-free interaction with IoT devices such as security doors and connected cars, without even having a need to take the smartphone out of the pocket. The Samsung Galaxy Note 20 Ultra embarks the first UWB-enabled Android device leveraging a variety of applications and services for smartphones. The integration of NXP's UWB chip will not only make the sharing of files easier but would also offer superior location tracking capabilities, allowing the phone to position the user even in a crowded area with numerous signals and obstacles such as walls or others.

Q. How will interoperability among different vendors be ensured?

A. To ensure interoperability between multiple vendors, various consortiums like the UWB Alliance, FiRa Consortium, and Car Connectivity Consortium have been working to achieve the common goal of creating a UWB-enabled ecosystem. The collaboration will further promote growth and a favourable regulatory environment—only to further fuel greater efficiencies for consumers. Recently, the UWB Alliance and FiRa Consortium established a joint liaison agreement to align their areas of focus and agree upon a way to work together on areas of common interest.

Q. What will 5G adoption in India mean for UWB?

A. 5G adoption in India will serve as a catalyst for the digital and industrial revolution, increasing opportunities across sectors in the industry through digitisation by the introduction of new services and products. UWB enables greater data throughput and better location precision for high-speed wireless network applications. In our increasingly connected world, UWB will provide opportunities for better and faster wireless communication. UWB will upscale the network technology as it has the ability to deliver such extensive bandwidth at incredibly high speeds with ultra-low latency.

Q. How can UWB technology be leveraged in times of Covid-19 pandemic?

A. As the country reopens itself following the unlock guidelines by the government, UWB technology and its capabilities can be leveraged fruitfully for everyone for contact tracing and social distancing. UWB can help in tracking and monitoring people who are under isolation. In addition to that, it can also help us alert about someone positive being present within the 183cm (6ft) range. **EFY**